

ADRIEN STAL

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EDUCATION

University of Toronto

Bachelor of Applied Science (BASc) + PEY Co-op, Electrical and Computer Engineering

Toronto, ON

2024 – 2028 (Expected)

- Available for 12–16 month co-op/PEY term starting May 2027

TECHNICAL SKILLS

Hardware & Verification: Verilog, SystemVerilog, ModelSim, FPGA (DE1-SoC), RTL Design, Digital/Logic Circuits, Datapath/FSM Design, Computer Architecture

Software & Scripting: C, C++, Python, PyTorch, Machine Learning, Linux/Unix, Shell Scripting, Assembly

Developer Tools: Intel Quartus Prime, Altium Designer, LTspice, Git, VS Code, Intel Pin

Tools/Equipment: Oscilloscope, function generator, bench DMM, DC bench supply, soldering iron, reflow oven

Spoken Languages: English (Fluent), French (Fluent), Chinese (HSK3), German (B2)

TECHNICAL PROJECTS

Reconfigurable Systolic Array Accelerator | *SystemVerilog, RV32I, MMIO, DSP* 2026

- Designed a reconfigurable **systolic array datapath** for **neural network matrix multiplication** in **SystemVerilog**, using multiplexers to route operands and partial sums, which enabled **run-time switching** between **weight, input and output stationary dataflows**.
- Engineered **precision switching Multiply Accumulate units** using **sub-word parallelism** on variable precision **DSP blocks**, **doubling parallel computational throughput** by transitioning from single 32-bit operations to dual 16-bit operations within a fixed budget of **174 hardware multipliers**.
- Built a **single cycle RV32I soft core** in **SystemVerilog** as a primary hardware controller and integrated the accelerator over **memory-mapped I/O**, enabling datapath, precision, and matrix dimensions to be set at run time from software using only standard load and store instructions.

Connect-Four Hardware Game (DE1-SoC) | *Verilog, ModelSim, VGA, PS/2, FSM* 2025

- Implemented **RTL design** in **Verilog** for a complete hardware-based logic system, integrating **datapath routing** for win-state detection and illegal-move validation.
- Designed complex **Finite State Machines (FSMs)** to orchestrate the main system control path, synchronizing **PS/2** hardware peripheral inputs with real-time **VGA rendering**.
- Utilized on-chip **Block RAM** to manage tile state data, successfully interfacing memory structures with a custom **VGA controller** to ensure pixel-accurate display synchronization.
- Verified digital logic and timing behavior using **ModelSim testbenches** prior to physical synthesis and routing on the **Altera DE1-SoC FPGA**.

Real-Time Embedded Audio Mixer (DE1-SoC) | *C, RISC-V, MMIO, ISRs, VGA* 2026

- Engineered a **real-time embedded audio mixing system** in **C**, utilizing **memory-mapped I/O** and **Interrupt Service Routines (ISRs)** to interface directly with hardware peripherals, enabling zero-latency user control.
- Processed and mixed large scale audio arrays in real time, interfacing directly with memory-mapped **hardware FIFO controllers** to monitor write space availability, preventing **buffer overflow** and ensuring synchronized **dual-channel audio output**.
- Designed a synchronized **VGA display controller** interface using direct **pixel buffer manipulation** and **double-buffering**, polling hardware **Vsync registers** which ensured real time and tear free UI rendering.

Multi-Head CNN Image Classifier (APS360) | *Python, PyTorch, ResNet-18, CNN* 2026

- Architected a **multi-head CNN** using a pretrained **ResNet-18** backbone, strategically freezing initial convolutional layers (2.8M parameters) and fine-tuning the final layer (8.6M trainable parameters) to optimize compute resources and prevent overfitting.
- Designed the network datapath to compress feature maps via **global average pooling** into a 512 dimensional embedding, routing data into **three parallel MLP classification heads** utilizing 128 node hidden layers, ReLU activations, and dropout logic.
- Developed a memory efficient **data processing pipeline** to extract, augment, and discretize continuous OpenStreetMap metrics, enabling the CNN to achieve a **71.4% test accuracy** on held-out data and significantly outperform traditional **Random Forest baselines**.